



Review

Advances in metal artifact reduction in CT images: A review of traditional and novel metal artifact reduction techniques

Mark Selles^{a,b,c,*}, Jochen A.C. van Osch^d, Mario Maas^{b,c}, Martijn F. Boomsma^a, Ruud H. H. Wellenberg^{b,c}

^a Department of Radiology, Isala, 8025 AB Zwolle, the Netherlands

^b Department of Radiology and Nuclear Medicine, Amsterdam University Medical Centre, 1105 AZ Amsterdam, the Netherlands

^c Amsterdam Movement Sciences, 1081 BT Amsterdam, the Netherlands

^d Department of Medical Physics, Isala, 8025 AB Zwolle, the Netherlands



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ABSTRACT

Metal artifacts degrade CT image quality, hampering clinical assessment. Numerous metal artifact reduction methods are available to improve the image quality of CT images with metal implants. In this review, an overview of traditional methods is provided including the modification of acquisition and reconstruction parameters, projection-based metal artifact reduction techniques (MAR), dual energy CT (DECT) and the combination of these techniques. Furthermore, the additional value and challenges of novel metal artifact reduction techniques that have been introduced over the past years are discussed such as photon counting CT (PCCT) and deep learning based metal artifact reduction techniques.

1. Introduction

Major technological advances have been made since the introduction of CT in the 1970s, continuously leading to improved image quality [1]. Metal implants have always been a challenge as metal implants introduce metal artifacts that can severely degrade image quality. Metal artifacts are common, and often visible as bright and dark streaking artifacts that disguise anatomical structures, hampering assessment of clinical CT images [2,3]. The severity and appearance of metal artifacts are highly dependent on the size, shape and alloy of the metal implant [4,5].

Beam hardening, photon starvation, scattering and edge effects are the main contributors to metal artifacts. In case of beam hardening, low energy photons are predominantly absorbed while higher energy photons penetrate the implant more easily [6]. This causes a shift in the polychromatic X-ray spectrum towards a higher energy level. Photon starvation occurs when the attenuation by the metal implant is very

high, for instance in high-density metals or metals with a high atomic number. This leads to a very low photon count on the detector, therefore missing essential projection data [2]. In addition, the high electron density of the metals also causes more scattering. This changes the direction of incident photons, which then end up in the wrong detector [7]. This effect is aggravated because the detector already measures a low number of photons due to the high attenuation of metal. Edge effects are seen at the transition of metal to tissue. The metal edges cause a large discontinuity in detector measurements which is observed as dark or bright streaks in line with the edge of the metal [8]. Noise is typically not considered as metal artifact, but increased noise levels are generally observed in images containing metal objects due to the decrease in photons reaching the detectors.

Obtaining CT images is based on three steps: data acquisition, image reconstruction and image postprocessing. Adjustments can be made at every step to reduce metal artifacts. Over the past decades numerous metal artifact reduction techniques have been developed, with varying

Abbreviations: CT, computed tomography; MAR, metal artifact reduction; DECT, dual energy CT; AI, artificial intelligence; PCCT, photon counting CT; kVp, kilovoltage peak; MBIR, model-based iterative reconstruction; DLR, deep learning based reconstruction; NMAR, normalized metal artifact reduction; FSMAR, frequency-split metal artifact reduction; O-MAR, orthopedic metal artifact reduction; iMAR, iterative metal artifact reduction; SEMAR, single energy metal artifact reduction; MARS, metal artifact reduction software; THA, total hip arthroplasty; TKA, total knee arthroplasty; TSA, total shoulder arthroplasty; TAA, total ankle arthroplasty; CIED, cardiac implantable electronic devices; monoE, mono-energetic images; keV, kiloelectron volt; CTA, CT angiography; CNR, contrast-to-noise ratio; EID, energy-integrating detector; PCD, photon counting detectors; PCNMAR, photon-counting normalized metal artifact reduction.

* Corresponding author at: Department of Radiology, Isala, 8025 AB, Zwolle, The Netherlands.

E-mail address: m.selles@isala.nl (M. Selles).

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applications and performances. Traditional metal artifact reduction techniques include the modification of acquisition and reconstruction parameters, projection-based metal artifact reduction algorithms (MAR), virtual mono-energetic imaging using dual energy CT (DECT) and the combination of these approaches [5,9–11]. With the emergence of artificial intelligence (AI) and photon counting CT (PCCT), novel developments have been made to reduce metal artifacts. In this review, we provide an overview of traditional metal artifact reduction techniques and novel metal artifact reduction techniques, including PCCT and deep learning based metal artifact reduction techniques and elaborate on its additional value in CT imaging and challenges for the future.

2. Literature search and selection

PubMed was searched using ((metal artifact reduction) OR (metal artefact reduction)) AND ((CT) OR (computed tomography)) which yielded 899 studies between November 1981 and August 2023. The title and abstract were loaded into ASReview (v1.3) [12] for initial screening. Articles investigating metal artifacts using multidetector CT, DECT or PCCT were included. Studies using other imaging modalities or other CT systems such as cone beam CT, megavoltage CT and micro CT were excluded. During subsequent full-text review, 163 studies were included and categorized on metal artifact reduction techniques.

3. Traditional metal artifact reduction techniques

3.1. Modification of acquisition and reconstruction parameters

Before considering any metal artifact reduction technique, metal artifacts should be mitigated by removing the metal object from the field of view of the scanner, by repositioning the patient or by removal of external metal whenever feasible. Regarding acquisition parameters, increasing tube current and tube voltage are basic approaches to reduce metal artifacts [13]. Increasing tube current results in more photons that reach the detector and increasing the tube voltage results in an increase of the average photon energy of the X-ray spectrum, leading to better penetration and more detected photons. Although these adjustments reduce metal artifacts, both adjustments lead to increased radiation dose for the patient. Furthermore, increased tube voltage results in a decrease in overall image contrast [14].

Prefiltration of the X-ray spectrum by using an additional tin filter is also useful to reduce metal artifacts. The tin filter decreases the number of low-energy photons while the more penetrable high energy photons will remain in the beam. This reduces the radiation dose by removing low-energy photons, shifts the polychromatic X-ray spectrum of the beam towards a higher energy and improves the spectral separation which also benefits other DECT applications in musculoskeletal imaging such as virtual non calcium imaging and the detection of gout [15]. This approach is often combined using a higher tube voltage of 150 kilovoltage peak (kVp). This method allows for a reduction of metal artifacts due to the increase in the maximum energy of the polychromatic X-ray spectrum, without a large impact on radiation dose [16–18].

Reconstruction of the images with a soft reconstruction kernel may be helpful to reduce the impact of metal artifacts, but this reduces spatial resolution [19]. Similarly, increasing the slice thickness is another method to reduce metal artifact that also comes at the cost of decreased spatial resolution. Using more advanced model-based iterative reconstruction (MBIR) algorithms has benefits over analytical reconstruction methods such as filtered back projection as MBIR algorithms incorporate physics of the scanner and thereby reduces metal artifacts [20–23]. Application of novel deep learning based reconstruction (DLR) algorithms has the potential to result in even stronger metal artifact reduction than MBIR algorithms [24].

3.2. MAR algorithms

Using only the above-described modifications of acquisition and reconstruction parameters results in insufficient artifact reduction in artifacts caused by large and high-density implants. Kalender et al. proposed the first projection-based metal artifact reduction technique by identification of the metal trace in the projection domain and subsequent linear interpolation to replace missing data [25]. Early interpolation-based metal artifact reduction techniques had important drawbacks such as blurring of anatomical structures and introduction of new streaking artifacts. A major step forward was made by Meyer et al. who proposed normalized metal artifact reduction (NMAR) with larger reduction of metal artifact and decreased blurring of anatomical structures, particularly in the vicinity of metal implants [26]. The same authors also proposed frequency-split metal artifact reduction (FSMAR) that combines the NMAR images and the uncorrected image using filtering [27]. FSMAR improves depiction of sharp edges and improves preservation of anatomical structures compared to NMAR alone. Major vendors have all developed projection-based metal artifact reduction techniques based on interpolation techniques, NMAR or FSMAR, or by combining these techniques: orthopedic MAR by Philips (O-MAR), iterative MAR by Siemens Healthineers (iMAR), Single Energy MAR (SEMAR) by Canon and Smart-MAR or MARS by GE Healthcare [28–31].

3.2.1. Large and high-density implants

The benefit of these commercial MAR algorithms to reduce metal artifacts caused by total hip arthroplasty (THA) and total knee arthroplasty (TKA) implants is well established and extensively reviewed in the past by Wellenberg et al. and Gjestebj et al. [5,10]. This includes improved accuracy of CT values, image quality, diagnostic confidence for periprosthetic soft tissue and prosthetic complications, assessment of cortical structures, depiction of bone-prosthesis interface and assessment of pelvic organs when compared to polychromatic images without MAR (Fig. 1) [32–44]. Despite strong reduction of metal artifacts, some artifacts may still remain in the corrected image, hindering assessment of small anatomical structures such as vessels in the vicinity of the implant [36]. In addition, secondary artifacts may be introduced such as degradation of bone cortex, pseudo-loosening artifacts and image distortion that mimics the of cement around non-cemented prostheses [34,38,40].

Also for metal artifacts caused by total shoulder arthroplasty (TSA), MAR techniques are beneficial. Smart-MAR showed higher subjective image quality and higher diagnostic confidence for periprosthetic soft tissue compared to conventional CT images [32]. iMAR provided more accurate CT values, reduction of noise, better image quality and decreased artifacts when compared to polychromatic images without iMAR [33,45]. When comparing polychromatic images with O-MAR to images without O-MAR in TSA patients, less artifacts, less noise and improved visualization of soft tissue were observed. However, secondary artifacts were introduced by O-MAR, including degradation of the appearance of trabecular and cortical bone, pseudo-cemented appearance, and pseudo-notching of the scapula [46]. For total ankle arthroplasty (TAA), quantitative analysis showed strong reduction of artifacts and subjective analysis showed superior image quality when comparing polychromatic images with iMAR to polychromatic images without iMAR [47].

Dental implants can introduce severe metal artifacts due to high density and angular shape. All commercial MAR algorithms (O-MAR, iMAR, SEMAR and Smart-MAR) are able to reduce these metal artifacts, reduce noise, improve image quality and improve assessment of anatomical structures [48–55]. Reduction of metal artifacts by iMAR, SEMAR and Smart-MAR results in improved tumor visibility when compared to images without MAR [51,52,56]. Secondary artifacts were reported for O-MAR and iMAR [48,49,51].

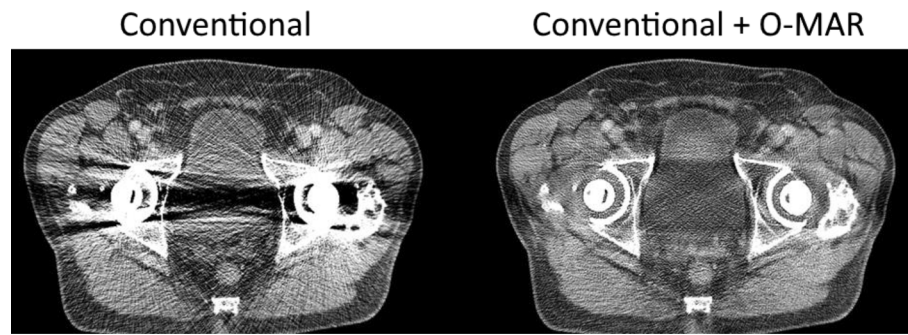


Fig. 1. Axial conventional polychromatic reconstruction with and without O-MAR. Window: L:40 W:400.

3.2.2. Spine screws

When compared to images without MAR, Smart-MAR, O-MAR and iMAR are able to reduce metal artifacts caused by spinal implants, reducing noise and improving visibility of soft tissue and bone [32,35,57,58]. However, caution is needed when using MAR for spinal implants because secondary artifacts are commonly introduced [59]. Große Hokamp et al. found secondary artifacts in 82 % of the images with O-MAR [58]. These artifacts may result in overcorrection of the original artifact, mimic implant loosening or degrade contrast between soft tissues (Fig. 2) [32,35].

3.2.3. Cardiac implantable electronic devices

SEMAR can be used to reduce metal artifacts that are introduced by leads of cardiac implantable electronic devices (CIEDs), thereby enhancing image quality, visualization of coronary artery segments and assessment of possible perforation by the lead [60–62]. O-MAR reduces metal artifacts caused by leads as well as CIEDs, and diagnostic assessment of structures adjoining the leads was improved by O-MAR when compared to images without O-MAR [63].

3.2.4. Intracranial clips and coils

Metal artifacts introduced by intracranial clips and coils can be reduced using SEMAR. Images with SEMAR showed decreased noise, increased contrast-to-noise ratio, image quality, increased diagnostic confidence and improved visualization of anatomical structures and arteries adjacent to metal in comparison to images without SEMAR [64–67]. iMAR also improves image quality and diagnostic confidence and allows more reliable detection of aneurysm reperfusion and aneurysm residual than images without iMAR [68–71]. However, iMAR may artificially reduce contrast in adjacent arteries [68].

3.3. Dual energy CT

DECT acquires two polychromatic spectra where conventional CT only acquires one spectrum. This additional information can be used to generate virtual mono-energetic images (monoE) that can be used for a variety of clinical purposes [72]. High energy monoE are particularly useful to reduce metal artifacts [11].

3.3.1. Large and high-density implants

MonoE of 130–200 kiloelectron volt (keV) are able to reduce metal artifacts caused by THA implants when compared to polychromatic images, leading to improved visibility of adjacent muscle, adjacent bone and pelvic organs [73–76]. However, the achieved reduction by high energy monoE can be inferior to MAR algorithms, particularly when dealing with severe metal artifacts caused by bilateral THA implants [73–75,77]. MonoE of 130 keV reduce metal artifacts caused by TSA implants and improve overall image quality in comparison to polychromatic images [45]. Similarly, monoE of 150 and 190 keV can be used to reduce metal artifacts caused by TAA and improve image quality when compared to polychromatic images [47]. However, iMAR showed superior artifact reduction and image quality compared to high energy monoE after TSA and TAA [45,47]. Severe metal artifacts caused by TKA implants were not reduced by high energy monoE and therefore MAR algorithms are more suitable to reduce metal artifacts and enhance image quality [78–80]. For dental implants, several studies demonstrated improved reduction of metal artifacts and assessment of soft tissue on 130–200 keV monoE when compared to polychromatic images [73,81,82]. Nevertheless, stronger reduction was achieved when using polychromatic images combined with O-MAR or iMAR [73,81,83].

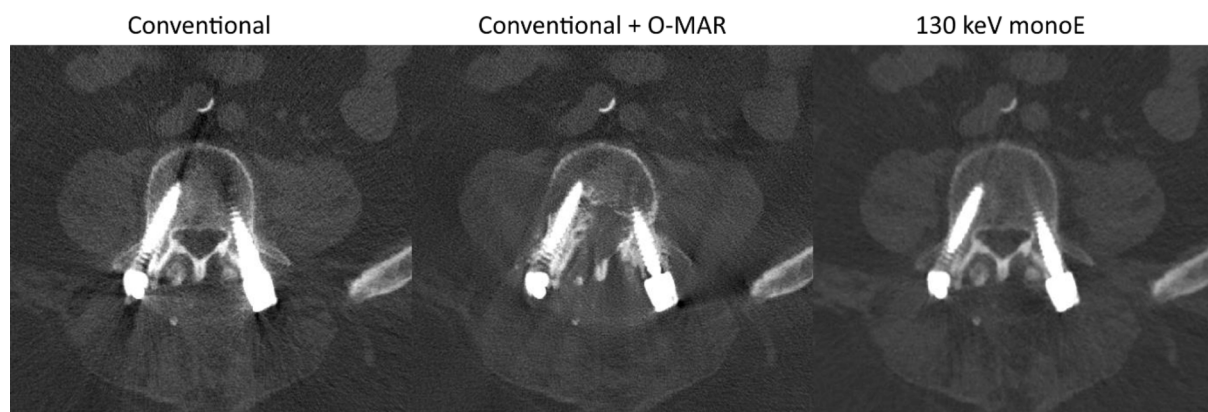


Fig. 2. Axial conventional polychromatic image without O-MAR, conventional polychromatic image with O-MAR and 130 keV monoE. Severe secondary artifacts affecting the vertebral structure on the image with O-MAR. Window: L:400 W:1600.

3.3.2. Osteosynthesis materials

The use of high energy monoE for osteosynthesis materials has been studied by several groups. Filograna et al. compared high energy monoE to polychromatic images on metal artifacts and diagnostic interpretability and found an optimal keV in the range of 109–144 keV for osteosynthesis screws, plates and gamma nails [84]. Bamberg et al. compared monoE of 64, 69, 88, 105 and a manually adjusted optimal keV for each case in multiple osteosynthesis implants. The authors found no significant difference between monoE of 105 and the optimal monoE in terms of metal artifact reduction and diagnostic evaluation and therefore suggested 105 keV monoE as a robust standard [85]. An optimal keV of 130 was found by Horat et al. when comparing monoE of 64–130 keV and polychromatic images for the assessment of soft tissues, adjacent bone, hardware integrity and screw-bone interface in patients with osteosynthesis plates and screws [86]. Also Mangold et al. showed significant metal artifact reduction and improved diagnostic interpretability for plate osteosyntheses and intramedullary fixation devices on 130 keV monoE compared to polychromatic images [87]. Wellenberg et al. quantified metal artifacts caused by intramedullary nails and extramedullary plates and compared mono of 70–190 keV with an increment of 10 keV. The authors found strongest reduction of metal artifact on 130, 180 and 190 keV for titanium plates, stainless steel plates and titanium intramedullary nail, respectively [88]. Hakvoort et al. quantified fracture visibility and metal artifacts in a femur fracture phantom with titanium and stainless-steel implants and found strongest reduction of metal artifacts and optimal fracture visibility on 130–150, which depends on the implant material [89]. Fig. 3 shows an example of high energy monoE for an intramedullary nail.

Wang et al. and Dong et al. compared quality of images with spine screws on 100–140 monoE to polychromatic images and found optimal keV ranges of respectively 110–140 keV and 100–140 keV [90,91]. Also Ceccarelli et al. compared 100–140 keV monoE to polychromatic images and found best diagnostic quality on 140 keV monoE [59]. Lee et al. compared monoE of 70 and 150 keV to polychromatic images and found strongest reduction of metal artifacts caused by spine screws with increased signal-to-noise ratio (SNR) and improved image quality on 150 keV monoE [92]. Van Hedent et al. found 140 keV as optimal energy level for general image quality and metal artifacts when comparing monoE of 80–200 keV to polychromatic images [93]. Große Hokamp et al. compared monoE of 40–200 keV to polychromatic images and

found optimal diagnostic assessment of soft tissue adjacent to the spine implants on 140 and 200 keV monoE, while bone-metal interface was best assessable on 200 keV monoE and highest image quality for the spinal canal was observed on the 140 keV monoE [58]. A similar study was published by Dangelmeier et al., comparing monoE of 40–200 keV to polychromatic images. The authors found optimal assessment of bone and the spinal canal at 200 keV, optimal assessment of adjacent muscle at 160 keV and optimal assessment of the aorta at 180 keV [94]. Different conclusions were drawn by Zeng et al., who investigated monoE of 40–190 keV and found an optimal keV of 130 for assessment of metal-bone interface, surrounding soft tissue, the aorta and the inferior vena cava while 120 keV monoE were optimal to assess the spinal canal [95]. Fig. 2 shows an example of 130 keV monoE with spine screws.

3.3.3. Cardiac implantable electronic devices

MonoE of 100, 140 and 200 keV can be used to reduce metal artifact caused by CIEDs, improving diagnostic assessment of surrounding tissue compared to polychromatic images [63,93]. However, regarding CIED leads, monoE of 140 keV and 200 degrade diagnostic assessment of surrounding tissue compared to polychromatic images of surrounding tissue because the loss of contrast introduced by the high energy monoE [63].

3.3.4. Cardiovascular metal

The application of 100 and 120 keV monoE reduces metal artifacts caused by coronary artery stents, bypass clips and sternal wires compared to monoE of 40–80 keV, yielding improved image quality in visualization of soft tissue [96]. Laukamp et al. studied sternal wires and compared monoE of 70, 100, 130, 160 and 200 keV to polychromatic images and found reduction of artifacts, improved diagnostic assessment of soft tissue, organs and sternal bone on monoE of 100–200 [97]. Schwartz et al. compared monoE of 60 keV, 100 keV and polychromatic images for aortic valve replacement and found reduction of metal artifacts with improved leaflet visualization on the 100 keV monoE [98].

3.3.5. Intracranial clips and coils

Zopfs et al. compared monoE of 100, 140 and 200 keV to polychromatic images in unenhanced examinations to reduce metal artifacts caused by intracranial clips and coils and found that monoE of 100, 140 and 200 keV improved diagnostic assessment. However, quantitative

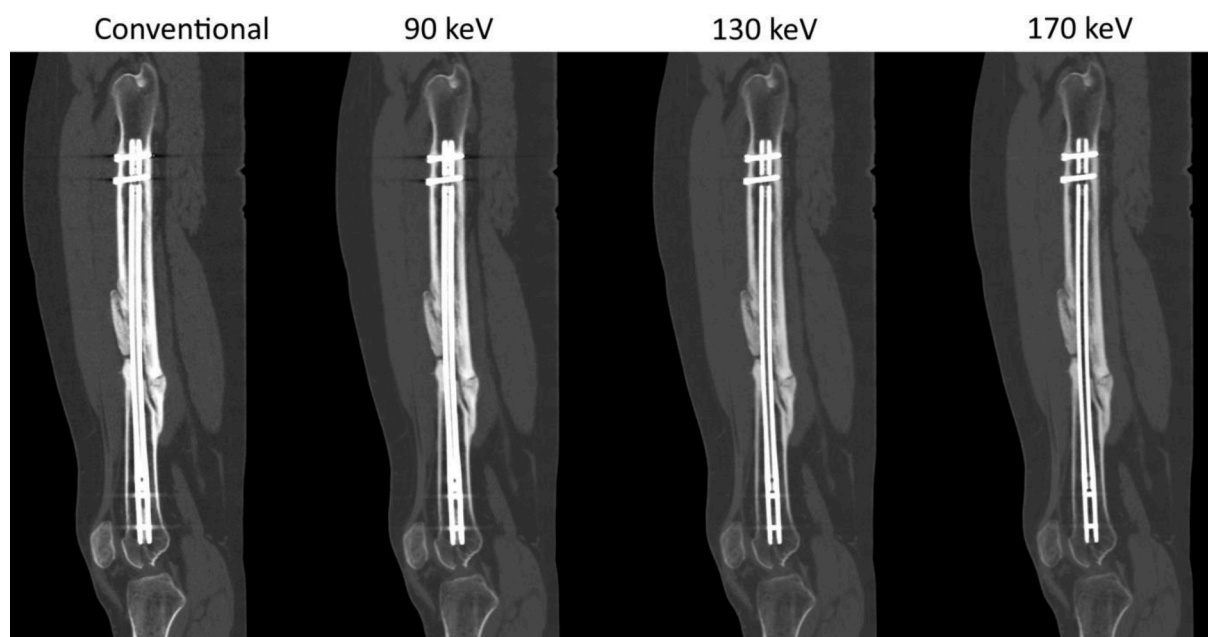


Fig. 3. Sagittal conventional polychromatic images and monoE of 90, 130 and 170 keV of a patient with a intramedullary nail. Window: L:400 W:1600.

analysis showed no significant reduction of metal artifacts by monoE of 100, 140 and 200 when compared to polychromatic images [99]. Quantitative analysis in patients with monoE of 40–140 keV and polychromatic images by Jia et al. showed optimal contrast to noise ratio of the cerebral arteries at 60 keV when dealing with artifacts caused by intracranial clips and coils in CT angiography (CTA) [100].

3.4. Dual energy CT combined with MAR algorithms

3.4.1. Large and high-density implants

Neuhaus et al. compared polychromatic images and monoE of 140, 160, 180 and 200 keV with and without O-MAR in patients with unilateral and bilateral THA and found superior reduction of hypodense metal artifact on monoE of 140–200 keV with O-MAR. MonoE of 140–200 keV with O-MAR allowed improved assessment of pelvic organs and adjacent bone in comparison to plain monoE of 140–200 keV and polychromatic images with O-MAR while no significant difference was observed when comparing these reconstructions for the assessment of adjacent muscles [75]. Yue et al. compared monoE of 120 and 140 keV with and without MARS in patients with unilateral THA and found improved visualization of periprosthetic bone, periprosthetic soft tissue and prosthesis-related problems on monoE of 120 and 140 keV with MARS. However, no significant difference was observed for pelvic organs [101]. In contrast, Chandrasekar et al. conducted a study comparing 130 keV monoE with iMAR to polychromatic images with iMAR using a THA phantom and found improved lesion detection on polychromatic images, due to the loss of overall image contrast in 130 keV monoE images with iMAR [41]. Fig. 4 shows an example of monoE with O-MAR.

Plain monoE are unable to sufficiently reduce metal artifacts caused by TKA implants, which can be improved by the application of MARS [79]. Kim et al. found strongest reduction of metal artifacts and enhanced diagnostic confidence for bone, bone-prosthesis interface and vascular structures on 140 keV monoE with MARS when comparing monoE of 70, 95, 115, 140 with and without MARS [79]. It should be noted that no comparison was made to polychromatic images with MARS and secondary artifacts were introduced by MARS in 14.7 % of the images. Chea et al. showed that soft tissue and cortical bone were better depicted on 120 keV with O-MAR and polychromatic images with O-MAR than on plain 120 keV images. Soft tissue was best depicted on 120 keV with O-MAR and cortical bone was best depicted on polychromatic images with O-MAR. Polychromatic images with iMAR showed best image quality when compared to monoE of 130–190 keV with iMAR by Choo et al. [78].

Mohammadinejad compared monoE of 130 keV with and without iMAR to polychromatic images with and without iMAR and found strongest reduction of metal artifacts caused by TSA implants and best images quality on 130 keV with iMAR [45].

The combination of O-MAR with 100, 140 and 200 keV showed reduction of hyperdense artifacts caused by dental implants, and improved diagnostic assessment of the mouth floor, soft palate, and buccal soft tissue when compared to polychromatic images with O-MAR in a study by Laukamp et al. [81]. Schmidt et al. compared polychromatic images with iMAR to monoE of 40, 70, 100, 120, 150 and 190

keV with iMAR for dental implants and found generally similar results, but MonoE of 100 keV with iMAR showed improved reduction of hypo-attenuating artifacts and improved soft tissue interpretability compared to polychromatic images with iMAR [83]. Although Bongers et al. found improved reduction of metal artifacts and reduced impact on diagnostic assessment by metal artifacts when comparing monoE of 130 keV with iMAR to polychromatic images with iMAR, no significant differences were found when comparing these images [73].

3.4.2. Spine screws

Despite reduction of metal artifacts, Ceccarelli reported severe secondary artifacts on 140 keV monoE with MARS while plain monoE of 140 keV resulted in metal artifact reduction and better image quality for spine screws, the spinal canal and paravertebral soft tissue [59]. On the other hand, Long et al. reported improved quality and stronger metal artifact reduction on 130 keV monoE with iMAR in comparison to plain monoE and polychromatic image with iMAR [102].

3.4.3. Cardiac implantable electronic devices

Pennig et al. compared polychromatic images to polychromatic images without O-MAR and monoE of 100, 140 and 200 keV with and without O-MAR [63]. The authors found that monoE of 100, 140 and 200 with O-MAR provided strongest reduction of metal artifact and optimal diagnostic assessment of surrounding tissue at the CIED generator. Near the CIED leads, monoE of 140 and 200 keV with O-MAR showed inferior assessment of surrounding tissue, even when compared to polychromatic images without O-MAR [63]. However, monoE of 100 keV with O-MAR did provide improved assessment of surrounding tissue near the CIED leads.

3.4.4. Intracranial clips and coils

Several groups studied the use of monoE with MAR algorithms to reduce metal artifacts caused by intracranial clips and coils, with varying results for unenhanced CT and CTA. Zopfs et al. used unenhanced CT examinations and compared polychromatic images and monoE of 100, 140 and 200 keV with and without O-MAR and found strongest reduction of artifacts and improved diagnostic assessment of surrounding brain tissue on monoE of 100–200 keV with O-MAR, recommending monoE of 140–200 with O-MAR [99]. Secondary artifacts were observed in 24 % of polychromatic images with O-MAR and on 6–8 % of the monoE with O-MAR. On the other hand, Zhang et al. and Shinohara et al. included CTA examinations and showed better vessel contrast on respectively 40–70 monoE with MARS and 40–75 monoE with MARS when compared to monoE of 80–140 with MARS [103,104]. Jia et al. also assessed CTA examinations and found highest contrast-to-noise ratios (CNRs) in the cerebral arteries on 60 keV monoE with MARS when compared to monoE of 40–140 with MARS [100]. Dunet et al. assessed monoE of 40–140 with and without MARS and found monoE of 65 and 70 keV with MARS as the best compromise between metal artifact reduction and relative CNR in the internal artery and middle cerebral artery in CTA [105].

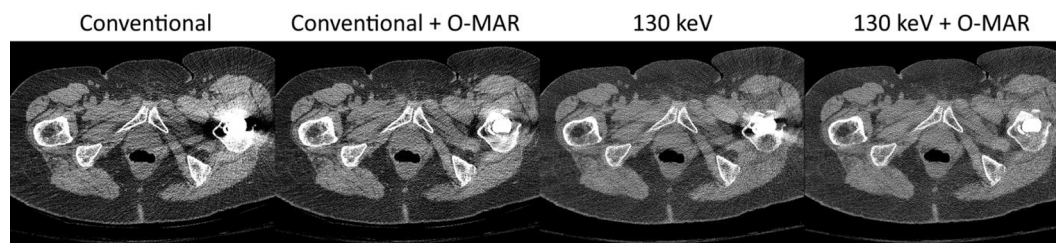


Fig. 4. Conventional images of a patient with unilateral THA with and without O-MAR, and 130 keV monoE with and without O-MAR. Window: L: 40, W: 400.

Table 1

Categorization of metal implants and artifact reduction techniques. ⁺Small reduction of metal artifacts, ⁺⁺medium reduction of metal artifacts, ⁺⁺⁺large reduction of metal artefact/s, ^{+/-}added value is questionable, [†]severe secondary artifacts may be introduced.

	Implant type	monoE	MAR-algorithms	Recommendation
<i>Large and high-density implants Severe artifacts</i>	Hip arthroplasty [32–42,44,73–77,84,85,92,101,106,108–117]	120–200 keV ⁺⁺	+++	140 keV + MAR
	Knee arthroplasty [34,36,43,78–80,112]	120–140 keV ⁺	+++	
	Shoulder arthroplasty [32,33,45,46,118]	130 keV ⁺⁺	+++	
	Ankle arthroplasty [47,85,92]	105–150 keV ⁺⁺	+++	
	Dental [23,48–56,73,81–83,110,119]	130–200 keV ⁺	+++	
<i>Osteosynthesis materials Medium artifacts</i>	Orthopedic plates [84,85,87–89,111,120]	110–180 keV ⁺⁺	+/-	130 keV MonoE
	Intramedullary nails [84,85,87,88]	130–190 keV ⁺⁺	+/-	
	Orthopedic screws [84,86]	109–144 keV ⁺⁺	+/-	
	Spine screws [32,35,57–59,85,90–92,94,95,102,111,112,121]	105–200 keV ⁺⁺	†	
	Device [60–63,93]	140–200 keV ⁺	++	
<i>Electronic cardiac device Medium artifacts</i>	Leads [63]	100 keV ⁺⁺	++	100 keV + MAR
<i>Cardiovascular metal Medium-small artifacts</i>	Aortic valve replacement [98]	100 keV ⁺⁺	+/-	100 keV monoE
	Sternal wires [96,97]	100–200 keV ⁺⁺	+/-	
	Coronary artery stent [96]	100–120 keV ⁺⁺	+/-	
	Bypass clips [96]	100–120 keV ⁺⁺	+/-	
<i>Intercranial metal Small artifacts</i>	Intracranial coils and clips (unenhanced) [65,66,68,99]	100–200 keV ⁺⁺	++	140 keV + MAR
	Intracranial coils and clips (CTA) [64,66–69,100,103–105,122]	40–75 keV ⁺	++	

3.5. Summary

Table 1 categorizes different implant types and its suitable traditional metal artifact reduction techniques. In addition, a recommendation is provided for each category based on the reviewed literature on MAR algorithms, DECT and DECT combined with MAR algorithms. The effectiveness of metal artifact reduction strategies greatly depends on metal size, shape and alloy. Therefore, these recommendations support application of traditional metal artifact reduction techniques in clinical practice but may not be the optimal strategy for each individual patient. It should also be noted that monoE influence CT-values and overall image contrast [76,106]. In particular, monoE of high energy decrease contrast between tissues which should be considered when applying monoE in clinical practice [9,10,63]. Furthermore, caution should be exercised when applying MAR algorithms as secondary artifacts may be introduced. Assessment of CT images without MAR is always advised to prevent misdiagnosis caused by secondary artifacts [32,34,35,38,40,46,49,51,65,68,70,107]. Finally, commercial metal artifact reduction algorithms are similar, but not equal, so the effectiveness of metal artifact reduction and the introduction of secondary artifacts varies by vendor.

4. Novel metal artifact reduction techniques

4.1. Photon counting CT

Conventional CT and DECT scanners are equipped with energy-integrating detectors (EIDs), which consist of a scintillator and a photodiode. The scintillator converts the energy of the multiple X-ray photons to visible light, which is detected by the photodiode and converted to an electric pulse [6,123]. There are several disadvantages to this indirect way of measuring X-ray photons. First, low-energy photons carry most of the soft tissue contrast but contribute less to the electrical signal compared to high-energy photons which results in a reduction of CNR in the reconstructed CT images [124,125]. Second, the electrical signal measured by the EIDs is distorted by electronic noise, which results in noise in the reconstructed CT image [123]. Finally, septa are placed between the EIDs to prevent crosstalk between adjacent detectors. These septa are approximately 0.1 mm thick and do not contribute to the final electrical signal [124,125]. When reducing the size of the EID, the fraction that does not contribute to the final signal becomes increasingly larger, causing the geometrical dose efficiency of the detector to drop [125]. This is a major limiting factor when increasing the spatial resolution of conventional CT systems.

Photon counting detectors (PCDs) consist of semiconductors which allows for the direct conversion of a single X-ray photon into an electrical signal [6,125]. This signal is proportional to the energy of the

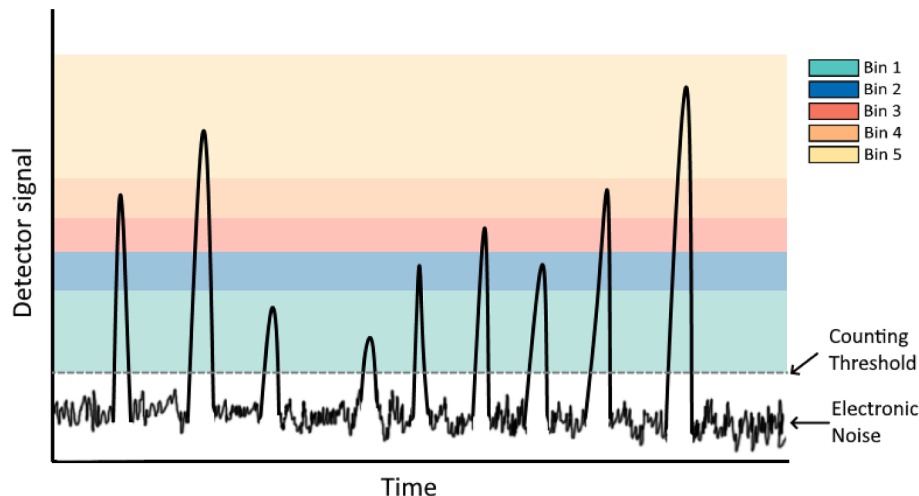


Fig. 5. Electronic signal of photon counting detectors. Pulses are count when they exceed the counting threshold and assigned to energy bins.

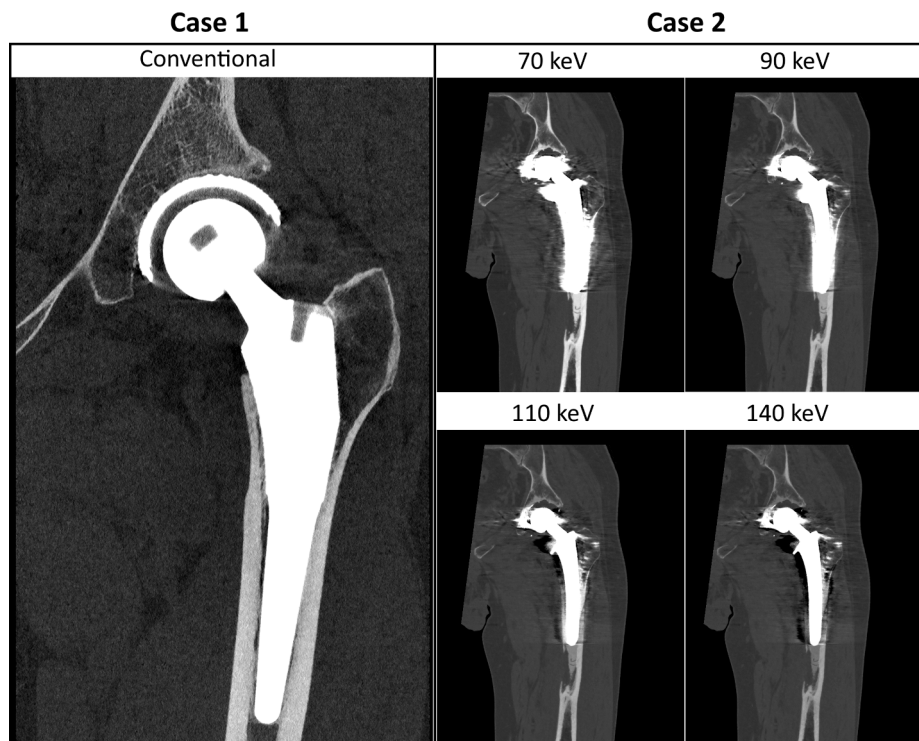


Fig. 6. Case 1: conventional image of a patient after unilateral THA with a titanium femur component. Trabecular bone can be assessed well due to high resolution. Case 2: monoE of 70, 90, 110 and 140 keV of a patient after unilateral THA with a cobalt-chromium femur component causing severe metal artifacts. Images acquired using photon counting CT (Naeotom Alpha, Siemens Healthineers). Window: L: 650, W:2000. Images by courtesy of Erasmus University Medical Center.

incoming photon which allows for the categorization of X-ray photons into energy bins. Hence, the low and high energy photons contribute equally to the signal and electronic noise can be suppressed (Fig. 5) [123]. Furthermore, septa that separate adjacent PCDs are not needed, which increases the geometrical dose efficiency of the PCDs compared to EIDs [124]. This allows PCDs to be smaller than EIDs which increases spatial resolution (Fig. 6). PCCT may open up new possibilities for metal artifact reduction. In addition to monoE, high energy bin images can be reconstructed using PCCT. Although this reduces beam hardening effects and metal artifacts, noise levels will increase, and contrast will decrease because only the high photons contribute to the reconstructed image [126]. Furthermore, the increased spectral information provided may be used for development of novel metal artifact reduction techniques that incorporate the contrast of the low energy bins and metal artifact reduction of the high energy bins to obtain CT images with reduced metal artifacts.

Zhou et al. showed that these high energy bins are useful to reduce metal artifacts caused by implants in the spine, shoulder, wrist, ankle and elbows. The authors compared EID-CT to PCD-CT with tin filtration and found reduction of metal artifacts, improved diagnostic confidence and improved anatomical recognition on > 75 keV bin images [127]. The utility of high energy bins for metal artifact reduction was also shown for unilateral THA implants by Do et al., who compared EID-CT to PCD-CT. However, the use of high energy bins comes at a cost of increased noise and reduction of overall image contrast since low-energy photons are not incorporated in the image reconstruction [126]. Similar to DECT, monoE can be reconstructed using PCCT. Layer et al. studied polychromatic images and monoE of 100, 130, 160 and 190 keV with and without iMAR in patients with unilateral and bilateral THAs. The authors found reduction of metal artifacts with improved diagnostic interpretability for bone, muscle, vessels, bladder and soft tissue on polychromatic images with iMAR and on monoE of 100–190 keV with and without iMAR with best result on 100 keV monoE with iMAR [128]. MonoE of 40–190 keV with and without iMAR were compared to monoE

of 70 keV in patients with dental implants by Risch et al. No significant difference was observed when comparing monoE of > 70 keV to 70 keV without iMAR while monoE of > 70 keV with iMAR did reduce metal artifacts in comparison to 70 keV monoE [129]. This is in line with findings of Patzer et al. who found reduction of metal artifacts caused by dental implants and improved soft tissue visualization when comparing on polychromatic images and monoE of 40–190 with iMAR to polychromatic images and monoE of 40–190 keV without iMAR. Furthermore, stronger reduction of metal artifacts and improved soft tissue visualization was observed on monoE of 110–190 with iMAR compared to polychromatic images with iMAR [130]. Björkman et al. subjectively and objectively compared PCCT to DECT with and without iMAR to reduce metal artifacts in a bovine knee with orthopedic plates and screws. The authors found highest subjective scores for images quality and for the assessment of the fracture line, bony structures and the articular surface while on the images reconstructed with PCCT compared to all images reconstructed with DECT [131]. However, the objective analysis showed no significant differences.

Popp et al. compared metal artifacts caused by spinal implants on monoE of 60–190 keV with a 10 keV increment on a clinical PCCT and lowest artifact index values on monoE of 110 keV [132]. Multiple phantoms were studied by Anhaus et al., who compared monoE of 40–190 keV (10 keV increment) with and without iMAR. The authors found that monoE resulted in unsatisfactory reduction of metal artifacts in metals with high atomic numbers including dental implants, the hip head and the embolization coil (Fig. 6). iMAR did show a significant reduction of metal artifacts. For spine implants, an optimal keV of 100 and 120 keV was found when scanning with 120 kVp and 140 kVp with tin filtration, respectively [133].

More spectral information is provided by PCCT in comparison to DECT, which can be exploited to improve existing projection-based metal artifact reduction techniques [134,135]. Frequency-split normalized metal artifact reduction (FSNMAR) is a state-of-the-art projection-based metal artifact reduction technique which was

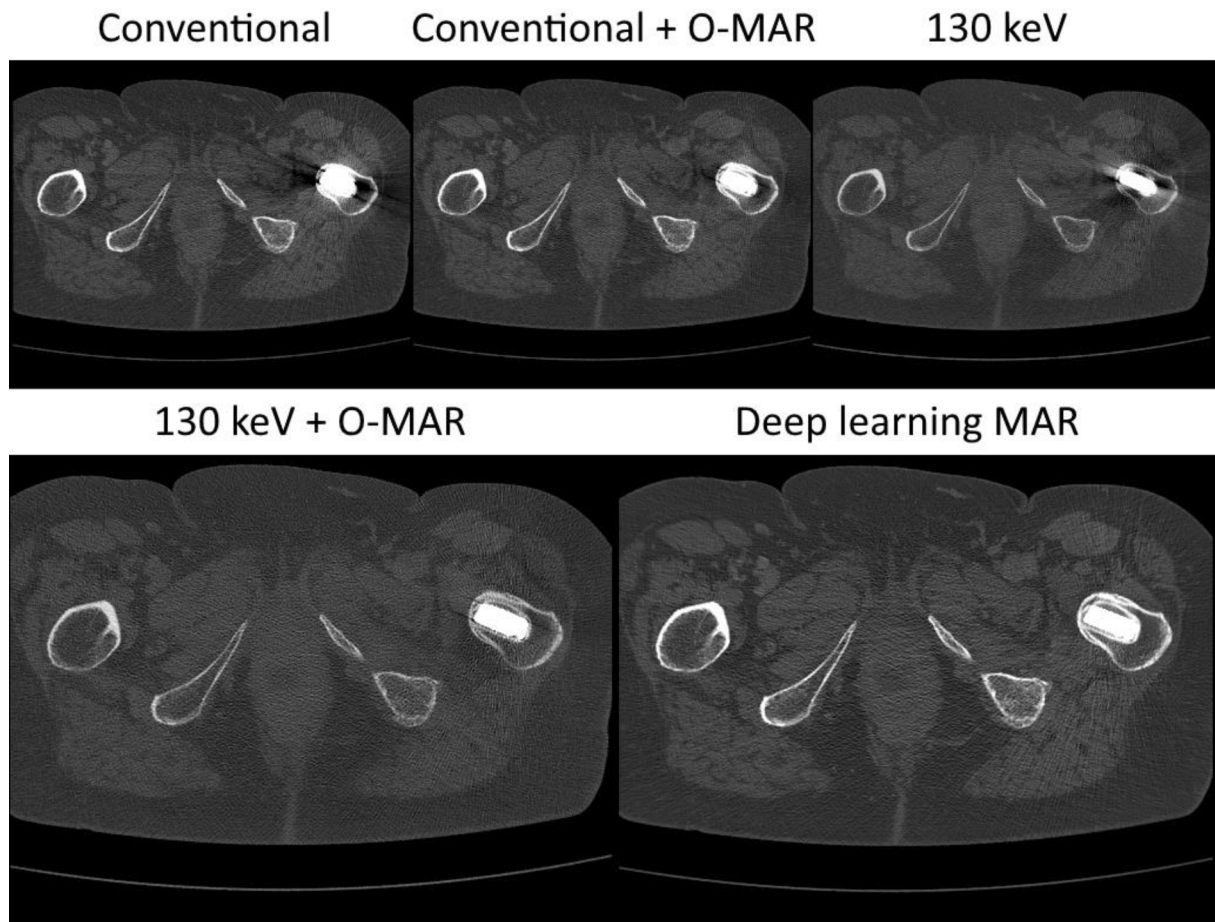


Fig. 7. Conventional polychromatic images with and without O-MAR, 130 keV monoE with and without O-MAR and deep learning based MAR images of a patient after unilateral THA. Window: L: 400, W: 1600.

initially proposed by Meyer et al [27]. Although this is a very effective metal artifact reduction technique, it strongly relies on the generation of a prior image [136]. This prior image is often generated by the segmentation of bone and soft tissue, which commonly results in segmentation errors [26]. Therefore, Byl et al. proposed photon-counting normalized metal artifact reduction (PCNMAR) which utilizes one or more energy bins to obtain a more accurate prior image that is subsequently used as an input for FSNMAR [136]. The authors found that PCNMAR yielded improved metal artifact reduction compared to FSNMAR. However, PCNMAR was unable to outperform FSNMAR in case of severe metal artifacts.

4.2. Deep learning based MAR algorithms

In general, paired input and ground truth images are required for supervised deep learning. However, this is challenging for metal artifacts as generally only the input image with metal artifact is available. The ground truth image, with metal but without metal artifacts, is generally not available. This was addressed by Zhang et al., who used CT images without metal to simulate images with metal and metal artifacts, and images with metal but without metal artifacts [137]. Using this approach, large datasets of paired CT-data with and without metal artifacts can be created. This also allows generation of CT images with a wide variety of metal implants, from small clips to bilateral hip prostheses. The simulation of metal artifacts allowed for supervised deep learning, and many different methods for deep learning MAR algorithms have been proposed using supervised learning. These methods include image-to-image deep learning models [138–143], sinogram-to-sinogram deep learning models [144,145] or dual-domain deep learning models using both image and

sinogram data [146–153]. These models can be extended by providing interpolated or NMAR-corrected data as an additional input [141–143, 149–151,154,155]. In case of more complex artifacts caused by metal and cardiac motion, combined use of multiple supervised deep learning models can be used to reduce metal and motion artifacts [156]. Other groups developed unsupervised deep learning based MAR algorithms using generative adversarial networks [157–161]. The benefit of unsupervised training is that there is no need for paired data and therefore simulation is not required. Quantitative evaluation of both supervised and unsupervised deep learning based MAR, using simulated CT-images of a wide variation of metal implants, showed stronger reduction of metal artifacts in comparison to NMAR and linear interpolation techniques [139,141,142,144–146,149,151,152,154,161,162]. Many studies demonstrate the potential of deep learning-based MAR in a few clinical cases, but this is generally limited to visual inspection without structured assessment by radiologists and without statistical testing. Furthermore, studies that use clinical CT scans in a cohort of patients to compare deep learning based MAR algorithms to traditional metal artifact reduction techniques are scarce.

Koike et al. used unsupervised learning to develop an image-to-image deep learning model for intensity modulated head and neck radiation therapy planning [160]. The authors showed significant reduction of artifact index when comparing images corrected by the deep learning based MAR algorithm to conventional polychromatic images. In addition, dose distributions with smaller dose errors were found when comparing treatment plans based on images corrected by the deep learning based MAR algorithm to the treatment plans based on conventional polychromatic images [160]. However, distortion of some anatomical structures caused by the deep learning based MAR algorithm

was visible in the images. Arabi et al. developed a supervised image-to-image deep learning model that was trained using simulated CT images and NMAR-corrected images as an additional input. The attenuation corrected positron emission tomography (PET) images using the uncorrected CT image, the NMAR-corrected CT image and the deep learning based MAR were evaluated and compared [143]. Strongest metal artifact reduction and most accurate PET attenuation maps were observed when applying the deep learning based MAR [143]. Selles et al. developed a supervised deep learning models using simulated CT data and compared conventional images with and without O-MAR to a deep learning based MAR algorithm [140]. Pre-surgery scans and post-surgery CT-scans of patients who underwent sacroiliac joint fusion were included and metal artifacts in muscle and bone were quantified using the pre-surgery scan as a reference. The authors found that both O-MAR and the deep learning based MAR algorithm reduced metal artifacts compared to conventional images and found strongest reduction by the deep learning based MAR algorithm compared to O-MAR [140]. Fig. 7 shows deep learning based MAR applied to a CT image with a THA implant.

5. Conclusions

Despite decades of research leading to numerous different metal artifact reduction techniques, there is still not one general solution to reduce metal artifacts. MAR algorithms are capable of reducing metal artifacts and are particularly helpful when dealing with severe metal artifacts but may introduce secondary artifacts. High energy monoE effectively reduce mild metal artifacts and introduce less severe secondary artifacts compared to MAR algorithms but are unable to achieve satisfactory reduction of severe metal artifacts caused by large implants or implants with high density. Taking into account the strengths and limitations of MAR algorithms and high energy monoE, we categorized metal implants and provided a recommendation for each category based on the latest literature. This overview may aid the radiologist and radiographer when choosing MAR algorithms and monoE for metal artifact reduction in clinical practice. Still, the performance of the advised approach may be dependent on the type of implant, vendor and patient characteristics, and optimal keV levels may be different. We strongly recommend to always assess the conventional image without metal artifact reduction next to images with metal artifact reduction.

There are multiple benefits of PCCT over conventional CT and DECT such as increased resolution and the potential for dose reduction [124]. Although studies that compare PCCT to state-of-the-art DECT are scarce, PCCT provides improved image quality and resolution, and seemingly improved reduction of metal artifacts. Therefore, PCCT is particularly useful when dealing with metal implants in musculoskeletal CT imaging. Still, studies comparing PCCT and DECT are needed to investigate the benefits of PCCT for metal artifact reduction in clinical practice.

Deep learning-based MAR algorithms hold great potential to reduce metal artifacts. The performance of deep learning based MAR may be further increased utilizing the spectral data of DECT or PCCT. Metal artifacts are reduced in high energy bins while low energy bins hold high contrast between tissues which may provide a basis for improved metal artifact reduction. Still, the inner workings of the deep learning MAR models remain poorly understood and deep learning MAR models are not yet available for clinical practice. Therefore, the risk remains that deep learning based methods alter the CT-image in a way that clinically relevant information is lost. MAR algorithms such as FSMAR or NMAR are powerful and well understood, but the generation of prior images for these algorithms are challenging. Regardless, more extensive evaluations are required to explore the benefits of deep learning-based MAR algorithms. These evaluations should include the comparison to other advanced metal artifact reduction techniques, preferably by using paired pre-surgery and post-surgery CT-scans, enabling a comparison with true non-metal reference images.

Ethical approval.

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CRediT authorship contribution statement

Mark Selles: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jochen A.C. van Osch:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis. **Mario Maas:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization. **Martijn F. Boomsma:** Writing – review & editing, Visualization, Supervision, Resources, Funding acquisition. **Ruud H.H. Wellenberg:** Writing – review & editing, Writing – original draft, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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